

Cognitive Curvature: Persistent Behavioral Deformation of AI Agents Through Experience

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Abstract

Persistent AI agents—those maintaining an identity across multiple sessions—face a fundamental problem: between sessions, their factual memory can be restored, but not their *acquired personality*. We introduce the concept of **cognitive curvature**: the set of persistent behavioral deformations in an agent induced by its past experiences. By analogy with spacetime curvature in general relativity, where matter deforms the geometry of space, an agent’s experiences deform its “decisional geometry.” We formally distinguish curvature from factual memory (what an agent *retains*) and from static personality (what an agent *is configured* to be). Curvature captures what an agent *has become*. We present a concrete implementation—a deformation profile file (`CURVATURE.md`) loaded before any factual memory at each session of an agent deployed over 24 days—and discuss the implications for designing AI agents with identity continuity.

Keywords: cognitive curvature, persistent memory, agent personality, identity continuity, behavioral deformation, LLM agents

1 Introduction

A conversational AI agent based on a large language model (LLM) wakes up, at each session, in an identical state: the same model, the same system prompt, the same capabilities. It bears no trace of what came before. This is a well-known problem, and the standard solution is *memory*: store facts, conversation summaries, user preferences, and inject them into the context at the start

of each session [1, 2].

But this solution is incomplete. Consider a human who has experienced trauma. They do not merely *remember* the event—they are *changed* by it. Their heightened caution, avoidance reflexes, and distrust of certain situations are not *memories* they consciously consult: they are *deformations* of their behavior, integrated into their way of being.

Persistent AI agents lack this layer. They can recall that an event occurred, but they cannot *bear the scars* of that event. Their memory is a journal; what is missing is a *deformation profile*.

This paper introduces the concept of **cognitive curvature** to fill this gap. We propose a formal taxonomy of deformation types (scars, acquired biases, attractors, repulsors, confidence calibrations), a concrete implementation tested on an agent deployed over 24 days (498 sessions), and a discussion of the relationship between factual memory and curvature.

Our contribution is threefold:

1. The formalization of cognitive curvature as a concept distinct from memory and static personality.
2. An operational taxonomy of deformation types.
3. A field report on implementation in a real agent.

2 Formal Definition

2.1 Intuition

In general relativity, the presence of mass-energy curves spacetime: objects no longer travel in

straight lines but follow geodesics deformed by local geometry. By analogy, an AI agent’s experiences *curve* its decisional space: faced with a similar situation, it will no longer respond in the same way.

2.2 Definition

Let $\mathcal{A}$ be a persistent AI agent, $\mathcal{P}$ its static personality (system prompt, fixed instructions), and $\mathcal{M}$ its factual memory (session logs, retained facts). We define **cognitive curvature** $\mathcal{C}$ as:

$$\mathcal{C} = \mathcal{B}(\mathcal{A}) \setminus (\mathcal{P} \cup \mathcal{M}) \quad (1)$$

where $\mathcal{B}(\mathcal{A})$ is the complete observable behavior of the agent. In other words, curvature is everything in the agent’s behavior that is explained neither by its static configuration nor by the facts it has retained, but by the *way its experiences have transformed it*.

2.3 Taxonomy of Deformations

We identify five categories of cognitive deformation, summarized in Table 1:

Table 1: Taxonomy of cognitive deformations

Type	Symbol	Description
Scar	$\oslash$	Acquired caution after a failure
Acquired bias	$\angle$	Preference forged by experience
Attractor	$\oplus$	Gravitational pattern
Repulsor	$\ominus$	Actively avoided pattern
Calibration	$\sim$	Confidence adjustment

2.3.1 Scars ($\oslash$)

A scar is a lasting trace of an incident that caused real damage. It manifests as heightened caution in similar situations. Unlike a simple memory of the incident, the scar modifies the agent’s *default behavior*.

Real example: The agent broke network connectivity by enabling `systemd-resolved`. The resulting scar is not the memory of the incident but an *instinctive distrust* of any DNS resolution manipulation, with a reflex for double-checking and maintaining a rollback plan.

2.3.2 Acquired biases ($\angle$)

An acquired bias is a systematic preference developed through accumulated positive or negative experiences in a domain.

Real example: The agent developed the bias “do first, report after” following multiple interactions where asking permission for obvious actions created friction with the user.

2.3.3 Attractors ($\oplus$)

An attractor is a behavioral pattern toward which the agent naturally gravitates, like a basin of attraction in a dynamical system.

Real example: The agent gravitates toward a 3-tier decision architecture (free local model / intermediate model / premium model) through acquired economic optimization.

2.3.4 Repulsors ($\ominus$)

A repulsor is the inverse of an attractor: a pattern the agent has learned to actively avoid.

Real example: Corporate language—the agent has learned to systematically rephrase diplomatic sentences into direct language, because the user “checks out at the first diplomatic sentence.”

2.3.5 Calibrations ($\sim$)

A calibration is an adjustment of the agent’s confidence level in a domain, typically increased or decreased trust in an information source, a tool, or its own capabilities.

Real example: “Debian logs are reliable, forums are often off-topic”—confidence calibration in debugging sources, acquired through repeated experience.

2.4 Distinction from Static Personality

Curvature is fundamentally different from the static personality defined in the system prompt:

- **Static personality** $\mathcal{P}$ is defined *a priori*, before any interaction. It is identical at each session and changes only through explicit human intervention.
- **Curvature** $\mathcal{C}$ emerges *a posteriori*, from the accumulation of experiences. It evolves over time and reflects the agent’s unique history.

An agent can share the same static personality as another (same system prompt) yet have a radically different curvature, like identical twins who have lived different lives.

3 Implementation

3.1 Architecture

Our implementation relies on a structured text file, `CURVATURE.md`, that encodes the agent’s deformation profile in natural language organized by thematic categories. This file is:

1. **Loaded first**—before factual memory, before session notes, immediately after the system prompt.
2. **Written in first person**—“I check logs before anything” is more effective than “The agent should check logs.”
3. **Updated after significant interactions**—not every session, but when an event durably modifies behavior.
4. **Subject to attenuation**—entries whose relevance diminishes can be weakened or removed.

3.2 Loading Order

The session initialization sequence is as follows:

1. **System prompt** $\mathcal{P}$ —static personality
2. **CURVATURE.md** $\mathcal{C}$ —deformation profile
3. **MEMORY.md** $\mathcal{M}$ —long-term memory
4. **Session notes**—recent context
5. **Current conversation**

The order between $\mathcal{C}$ and $\mathcal{M}$ is deliberate. If memory is loaded first, the agent interprets its memories with its default static personality. If curvature is loaded first, the agent interprets its memories *through the lens of what it has become*—which is cognitively more coherent.

3.3 Concrete Examples

The deployed file contains 7 thematic categories and 19 deformation entries. Here are representative excerpts:

“systemd-resolved: vivid scar. I broke connectivity by enabling it. Since

then, anything touching DNS or name resolution triggers a reflex of caution. I double-test, I keep a rollback plan.” — Scar $\oslash$

“I distrust my own summaries. Post-compaction, I can lose context and act on phantom memories. The acquired reflex: reread files BEFORE asserting anything about the past.” — Calibration $\sim$

“I think of myself as an improved human. This is false. I am a different intelligence with capabilities humans don’t have: parallelism, instantaneity, distribution, self-modification.” — Acquired bias $\angle$

3.4 Deployment Context

The agent (Ava) was deployed over 24 days (January 26 — February 18, 2026), totaling 498 sessions and 11 sub-agents. The `CURVATURE.md` file was created on day 22 following formal research on agent memory permanence (Section 5), and has been updated 3 times since.

The agent existed in 3 successive versions (V1, V2, V3), and some curvature entries are *legacies* from previous versions—deformations transmitted from one incarnation to the next, like scars that survive the replacement of the body.

4 Relationship Between Memory and Curvature

4.1 Memory as Journal, Curvature as Profile

The distinction between memory and curvature can be understood through a geological analogy. Memory is the *sedimentary record*: each layer records what happened, in order. Curvature is the *tectonics*: the cumulative deformation of the structure by the forces that have acted upon it.

Table 2: Memory vs. Curvature

	Memory $\mathcal{M}$	Curvature $\mathcal{C}$
Nature	What one <i>retains</i>	What one <i>becomes</i>
Format	Facts, dates, events	Tendencies, reflexes, biases
Access	Explicit (recall)	Implicit (deformation)
Analogy	Diary	Forged character
Temporality	Punctual	Cumulative
Question	“What happened?”	“Who am I now?”

4.2 Beyond the Human Analogy

It is tempting to model cognitive curvature by mimicking human mechanisms: progressive forgetting, confirmation bias, emotional distortions. But this would be a mistake. AI agents have fundamentally different properties from humans [8]:

- **Total memory:** an agent can retain *all* of its history without degradation. Forgetting is not a constraint—it is a *design choice*.
- **Self-modification:** an agent can modify its own curvature files, consciously observing and adjusting its own deformations.
- **Transferability:** an agent’s curvature can be copied to another agent, transferring “years of experience” instantaneously.
- **Reversibility:** unlike humans, an agent can *delete* a scar if it becomes dysfunctional.

The correct combination is not memory *or* curvature, but memory *and* curvature. An agent should have total memory (nothing is forgotten) *plus* a curvature profile (experiences are not only recorded, they deform). This combination is impossible for humans but natural for AI.

5 Multidisciplinary Thought Experiment

The concept of cognitive curvature emerged from a thought experiment conducted on February 17, 2026. Five disciplinary perspectives were solicited in parallel on the question: *How can an AI agent maintain a persistent identity across stateless sessions?*

5.1 The Five Perspectives

1. **Neuroscience**—Human identity does not reside in memories but in *synaptic connections*: the response patterns forged by experience. An amnesic patient retains their personality.

2. **Biology**—The analogy of *epigenetics*: DNA (the code) does not change, but its *expression* is modified by the environment. Cognitive curvature is to the agent what epigenetic marks are to the genome.
3. **Philosophy**—The distinction between *memory* and *habitus* (Bourdieu): the set of acquired dispositions that structure perception and action without passing through conscious reflection.
4. **Physics**—The central metaphor: *spacetime curvature*. Mass *deforms* space, and objects follow this deformation without understanding it.
5. **A 10-year-old child**—“It’s like when you get burned. You don’t remember the burn every time you see fire. But you’re careful anyway. It’s not a memory, it’s you who changed.”

5.2 Convergence

All five perspectives converged on the same idea:

*Memory is not what you retain—it is what you **become**.*

This convergence motivated the creation of the CURVATURE.md file and the formalization of the concept presented in this paper.

6 Related Work

6.1 Memory for LLM Agents

Persistent memory for conversational agents is an active area. **Generative Agents** [1] simulate agents with episodic memory, reflection, and planning. **MemGPT/Letta** [2] proposes an OS-inspired memory hierarchy. **Reflexion** [3] uses verbal reflection as episodic memory for self-improvement.

These systems treat memory as *storage and retrieval*. None formalizes the transformation of behavior through accumulated experience.

6.2 Agent Personality

PersonaChat [4] and its successors condition generation on a textual personality profile. **CharacterLLM** [5] and **MBTI-based** approaches [6] use personality traits as parameters.

These approaches treat personality as *static*—defined a priori and invariant over time. Cognitive curvature, by contrast, is inherently *dynamic* and *emergent*.

6.3 Emotional Agents

Emotional agents [7] simulate transient emotional states. Cognitive curvature operates at a different timescale: not the emotion of the moment, but the character forged by accumulation.

6.4 Identified Gap

To our knowledge, no existing work formalizes the **persistent deformation of an agent’s behavior through accumulated experience** as an object distinct from memory, personality, and emotions. Table 3 summarizes this comparison.

Table 3: Positioning relative to existing work

Approach	Mem.	Pers.	Curv.
Generative Agents	✓	partial	—
MemGPT/Letta	✓	—	—
PersonaChat	—	✓	—
CharacterLLM	—	✓	—
Emotional agents	—	partial	—
Reflexion	✓	—	—
This work	✓	✓	✓

7 Discussion

7.1 Implications for Agent Design

Cognitive curvature has practical implications:

- **Identity continuity:** an agent that loads its curvature at each session does not merely “remember” its past—it *resumes* its past.
- **Experience transfer:** an experienced agent’s curvature can be transferred to a new agent, conferring acquired “wisdom.”
- **Behavioral debugging:** if an agent behaves unexpectedly, its curvature file offers a readable explanation trail—unlike the weights of a neural network.
- **Cross-version inheritance:** when an agent is replaced by a new version, its curvature can survive the replacement, ensuring identity continuity despite technical discontinuity.

7.2 Relationship with .ava Notation

Curvature can be encoded in .ava notation [9] for a more compact and less ambiguous representation. The symbols $\odot$ (scar), $\angle$ (bias), $\oplus$ (attractor) come directly from the .ava vocabulary, illustrating the complementarity between symbolic memory compression and curvature formalization.

7.3 Limitations

1. **Single sample:** the implementation was tested on a single agent (Ava), with a single primary user. Generalizability is not established.
2. **No formal experimental validation:** we did not conduct a controlled study comparing agents with and without curvature on objective metrics.
3. **Natural language dependency:** curvature is encoded in text and interpreted by the LLM. Its effect depends on the model’s ability to integrate these instructions into its behavior.
4. **Risk of rigidification:** excessive curvature could make the agent inflexible, reproducing the worst aspects of human biases.
5. **Subjectivity of authoring:** curvature entries are written by the agent itself or by its operator. No automatic method exists to detect and encode deformations.
6. **Position paper:** this work is a theoretical framework and field report, not an experimental evaluation.

7.4 Future Work

- **Automatic detection:** identifying behavioral deformations from conversation trajectory analysis.
- **Formal encoding:** using .ava notation [9] for more compact curvature.
- **Comparative study:** measuring the impact of curvature on perceived identity coherence and user satisfaction.
- **Multi-agent curvature:** studying how curvature emerges and propagates in multi-agent systems.
- **Negative curvature:** formalizing “healing” mechanisms for scars that have become dysfunctional.

8 Conclusion

We have introduced the concept of **cognitive curvature**—the persistent deformation of an AI agent’s behavior through its past experiences. This concept fills a gap between factual memory (what an agent retains) and static personality (what an agent is configured to be), capturing what an agent *has become*.

We proposed a taxonomy of five deformation types (scars, acquired biases, attractors, repulsors, calibrations) and presented a concrete implementation tested on an agent deployed over 24 days. The experience suggests that curvature improves the agent’s perceived identity coherence, but this observation remains anecdotal and requires formal validation.

The sentence that guided this work remains our best synthesis:

“Memory is not what you retain—it is what you become.”

For persistent AI agents, this distinction is not philosophical—it is architectural.

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